



Replicating and Extending IMM-TSF
A Benchmarking Study on Irregular Multimodal
EPA-Air Quality Data

STAT 8240 Data mining II

A PROJECT REPORT

Submitted by

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1. Abstract

This project investigates the challenges of forecasting using messy real-world data that includes both numerical sensors and unstructured text. Using the IMM-TSF (Irregular Multimodal Multivariate Time-Series Forecasting) framework, we focus on the EPA-Air dataset. We have successfully replicated the baseline performance metrics from the original research, confirming that integrating textual data reduces prediction error. Then we propose an architectural extension to the modality fusion module to further enhance the model's ability to handle asynchronous data streams. Traditional time-series models often assume that data is collected at regular intervals and consists only of numbers. However, real-world environmental data (like our EPA-Air quality) is often irregular and multimodal containing sensor readings alongside text-based reports or location based metadata. This project utilizes the TIME-IMM benchmark library to bridge the gap between clean laboratory datasets and complex real-world deployment scenarios. Specifically, under Track A (Reproducibility), we successfully reproduced 7 of 11 baseline models on the EPA-Air dataset, achieving results closely aligned with the original paper and confirming an average MSE reduction of 2.1% when text is incorporated. Under Track C (Methodological Extension), we conducted three targeted ablations comparing GPT-2 and BERT text encoders, analyzing architecture family interactions, and running a placebo test with noise embeddings revealing that encoder choice and model architecture jointly determine whether textual context genuinely helps or harms forecasting performance.

2. Introduction

Time series forecasting is a fundamental task across a wide range of real-world domains, including environmental monitoring, healthcare, finance, and network management. Most classical forecasting methods and even many modern deep learning approaches are built on a simplifying assumption that observations arrive at fixed, regular intervals and consist purely of numerical measurements. While this assumption is convenient for model design, it rarely holds in practice. Real-world data collection systems are messy. Sensors fail, report at different rates, and produce observations that are sparse, uneven, and asynchronous across measurement streams. At the same time, many of these systems generate rich contextual information in the form of unstructured text clinical notes, maintenance logs, news articles, or weather reports that arrives on its own independent schedule and is rarely aligned with the numerical observations it describes.

The EPA-Air dataset that this project focuses on is a precise example of this real-world complexity. It consists of four environmental sensors AQI, Ozone, PM2.5, and Temperature monitored across eight U.S. counties. Each sensor operates on its own sampling schedule, meaning observations across features are inherently misaligned in time. This form of irregularity is formally classified as Multi-Source Asynchrony in the TIME-IMM taxonomy, and it represents one of the most structurally challenging irregularity types for multimodal forecasting models because there is no natural alignment point between numerical streams or between numerical and textual data. With a Feature Observability Entropy of just 0.3777 the lowest among all nine datasets in the TIME-IMM benchmark EPA-Air dataset exposes significant imbalance in how evenly features are observed, making it a demanding and realistic testbed.

This project we are replicates and extends the IMM-TSF framework specifically on the EPA-Air dataset, pursuing two complementary tracks.

Track A: Reproducibility aims to verify the central empirical claims of the paper by re-running the baseline experiments under the same configuration Adam optimizer with learning rate $1e-3$, batch size 8, early stopping on validation loss, a 7-day context window, and a 7-day forecast horizon, with data split chronologically into 60% training, 20% validation, and 20% test. We successfully reproduced 7 of the 11 baseline models DLinear, Informer, PatchTST, TimesNet, TimeMixer, TTM, and tPatchGNN. Four models could not be run. TimeLLM and the default tPatchGNN configuration failed because their default patch size of 24 is incompatible with EPA-Air's sparse feature structure: with Feature Observability Entropy of 0.3777, many features produce fewer observed points per window than the patch tokenizer requires, causing empty patch tensors that crash training. We resolved this for tPatchGNN by reducing patch size to 4. CRU, LatentODE, and NeuralFlow failed due to ODE solver instability on EPA-Air's fine-grained 1.02-hour mean inter-observation interval integrating continuous dynamics over a 7-day horizon at this resolution accumulates numerical errors that cause gradient explosion during early training. Our reproduced MSE values fall within ± 0.05 of the paper's reported values for most models, and the direction of text effect whether adding text helps or hurts a given model matches the paper for 6 of 7 models. Our average MSE reduction from text integration is 2.1%, compared to the paper's reported 6.71% average across all nine datasets, a difference explained by EPA-Air's weak text-to-sensor semantic alignment and our subset of models skewing toward patch-based architectures.

Track C: Methodological Extension probes the boundaries of the paper's architectural claims through three targeted ablations. The first replaces the default GPT-2 text encoder with BERT across all seven reproduced models, directly testing the given paper's results metrics finding that encoder choice does not strongly influence results. We find that on EPA-Air specifically, encoder choice produces performance swings of up to 13 percentage points for individual models, with BERT outperforming GPT-2 on 3 of 7 models most notably flipping TTM from text-harmful to text-beneficial. The second ablation classifies models by architecture family non-patch models (DLinear, Informer, TimesNet, TimeMixer) versus patch-based models (PatchTST, TTM, tPatchGNN) and examines whether text benefit is consistent within families across both encoders. We find a clean split non-patch models benefit from text with both encoders (average of 4.8% MSE), while patch-based models are harmed by text with both encoders, though BERT causes less damage than GPT-2. This architecture-dependent sensitivity is masked by the paper's cross-dataset, cross-model averaging. The third ablation is a placebo test in which real GPT-2 embeddings are replaced with Gaussian noise vectors that match the per-dimension mean and standard deviation of the real embeddings, so the model receives inputs of identical shape and distribution but carrying no semantic content. For 3 of 7 models, noise performs comparably or better than real text, indicating that the observed multimodal gain in those models reflects a regularization effect from the additional input rather than genuine semantic signal from the weather news summaries.

3. Problem formation

This project addresses the problem of forecasting future values of irregular, multimodal, multivariate time series data using IMM-TSF. Specifically, given a set of past numerical sensor observations recorded at irregular and asynchronous timestamps alongside a separate stream of timestamped text documents, the goal is to learn a model that integrates both modalities to predict sensor values over a future 7-day horizon. In the EPA-Air dataset, this means forecasting four environmental variables AQI, Ozone, PM2.5, and Temperature across eight U.S. counties, where each sensor reports on its own independent schedule and 1,244 weather news summaries arrive asynchronously with no forced alignment to the sensor readings. The core challenge is that standard forecasting models assume regular, unimodal inputs, so the problem requires handling missing observations, variable sampling rates across features, and the integration of a semantically misaligned text modality all simultaneously.

4. Dataset & It's Description

This project uses the EPA-Air dataset from the TIME-IMM benchmark, which represents the Multi-Source Asynchrony irregularity type under the Artifact-Based category of the TIME-IMM taxonomy. The dataset consists of four environmental sensor variables AQI, Ozone, PM2.5, and Temperature monitored across eight U.S. counties: Los Angeles (CA), Maricopa (AZ), Philadelphia (PA), Bexar (TX), Dallas (TX), Richmond (VA), Hillsborough (FL), and Denver (CO).

The numerical stream contains 49,552 total observations across 6,587 unique timestamps, with a mean inter-observation interval of 1.02 hours. Each sensor operates on its own independent reporting schedule, creating inherent asynchrony across features. This is reflected in the dataset's Feature Observability Entropy of 0.3777 the lowest among all nine TIME-IMM datasets indicating that some sensors are observed far less frequently than others within the same time window.

The textual modality consists of 1,244 weather-related news summaries collected from Common Crawl News, filtered to mention the county name and the keyword "weather," and summarized into five-sentence narratives.

For all experiments, data is split chronologically into 60% training, 20% validation, and 20% test sets, with a 7-day context window and a 7-day forecast horizon applied per entity (county). Table 1 summarizes the key dataset statistics.

Property	Value
Entities (counties)	8
Numerical features	4 (AQI, Ozone, PM2.5, Temp)
Total observations	49,552
Unique timestamps	6,587
Mean inter-observation interval	1.02 hours
Feature Observability Entropy	0.3777 (lowest in benchmark)
Text entries	1,244
Context window	7 days
Forecast horizon	7 days

Table 1: Dataset properties

5. Methodology

This section describes the full experimental methodology for both tracks of the project. Track A covers the reproducibility study in which we replicated published baseline results on the EPA-Air dataset using the IMM-TSF benchmark library. Track C covers three targeted ablation experiments that extend the multimodal fusion analysis by varying the text encoder, isolating architecture-family effects, and conducting a placebo test with noise embeddings.

5.1 Overview

Our methodology follows the IMM-TSF framework proposed by Chang et al. (NeurIPS 2025), applied specifically to the EPA-Air dataset. The pipeline consists of four sequential stages which are data pre-processing and canonical pre-alignment, text encoding, timestamp-to-text fusion (TTF), and multimodality fusion (MMF). We implement both Track A (Reproducibility) and Track C (Methodological Extension) within this unified pipeline, with Track C introducing targeted modifications at specific stages to probe the framework's architectural assumptions.

5.2 Track A: Reproduction Setup

Canonical Pre-Alignment: Since EPA-Air contains irregular and variably sampled sensor readings, standard forecasting architectures cannot directly consume the raw data. We apply the canonical pre-alignment strategy from t-PatchGNN to convert each irregular input window into a fixed-length tensor. For each feature, three channels are constructed the observed sensor value (or zero if missing), a binary missingness mask indicating whether the observation was truly recorded,

and a normalized timestamp. Future query timestamps are appended as unobserved rows, allowing the model to condition its predictions on when they are needed. This preserves the irregularity signal rather than erasing it through imputation.

Text Encoding: Weather news summaries from the EPA-Air text stream are encoded using a frozen pretrained language model. Following the paper's default configuration, we use GPT-2 as the text encoder in all Track A runs. The encoder processes each five-sentence weather summary and produces a 768-dimensional embedding vector. Since the encoder is frozen during training, all embeddings are precomputed once and cached in our implementation, stored as .pt files in Google Drive and restored at the start of each session to avoid recomputation. A learned linear projection maps encoder outputs to the shared 768-dimensional fusion space.

Timestamp-to-Text Fusion (TTF): Since text and sensor observations arrive asynchronously with no forced alignment and the TTF module bridges the two streams by producing a temporally aware text representation for each forecast query time. We use the Recency-Weighted Averaging (RecAvg) variant as the default, which computes a Gaussian-weighted average of all past text embeddings, assigning higher weight to text entries whose timestamps are closer to the forecast query time. This allows the model to emphasize the most temporally relevant context without requiring explicit alignment.

Multimodality Fusion (MMF): The aligned text representation from TTF is combined with the unimodal numerical forecast using the GRU-Gated Residual Addition (GR-Add) module. GR-Add concatenates the numerical forecast vector and the text embedding, passes them through a GRU to compute a residual correction, and blends this correction into the base forecast through a learned gating mechanism. The gate dynamically controls how much textual context influences the final prediction, allowing the model to suppress noisy or irrelevant text signals an important property for EPA-Air where weather news has weak semantic alignment with sensor readings.

Baseline Models and Training

We reproduce 7 of the 11 baseline models from the paper across three architecture families:

- **Regular TSF models:** DLinear, Informer, PatchTST, TimesNet, TimeMixer
- **Large TS model:** TTM
- **Irregular TSF model:** tPatchGNN (patch_size reduced from 24 to 4)

All models are trained using the Adam optimizer with learning rate 1e-3 and batch size 8, with early stopping based on validation loss and a patience of 50 epochs by enabling early stop feature. Each model is evaluated in both unimodal (numerical only) and multimodal (numerical + text) settings. MSE is the primary evaluation metric.

5.3 Track C: Methodological Extensions

We conduct three targeted ablations, each modifying one component of the IMM-TSF pipeline while keeping everything else fixed.

Ablation 1: Text Encoder Swap (GPT-2 vs BERT)

The paper claims in Figure 6c that encoder choice does not strongly influence forecasting results. To test this claim specifically on EPA-Air, we replace GPT-2 with BERT-base as the frozen text encoder across all 7 reproduced models. BERT embeddings are precomputed separately and cached to Google Drive in the same `.pt` format used for GPT-2. All other pipeline components TTF module, MMF module training configuration remain identical. This isolates the encoder as the only variable and allows a direct per-model comparison of GPT-2 versus BERT performance.

Ablation 2: Architecture Family $\times$ Encoder Interaction

Rather than averaging encoder effects across all models as the paper does, we classify our 7 models into two architecture families and analyze encoder sensitivity within each group. Non-patch models DLinear, Informer, TimesNet, TimeMixer process the full aligned input tensor without patch tokenization. Patch-based models PatchTST, TTM, tPatchGNN tokenize the input into fixed-length temporal patches before encoding. We compute the mean percentage change in MSE relative to the unimodal baseline for each encoder within each family, revealing whether encoder sensitivity is architecture-dependent rather than uniformly negligible.

Ablation 3: Placebo Test (Real Text vs Distribution-Matched Noise)

To determine whether observed multimodal gains reflect genuine semantic understanding or simply a regularization effect from adding an extra input, we replace real GPT-2 embeddings with Gaussian noise vectors that match the per-dimension mean and standard deviation of the real embeddings. The noise is generated with a fixed random seed (42) for reproducibility. Since the noise tensors have identical shape and distribution to the real embeddings, any performance difference between the real-text and noise conditions can be attributed to semantic content rather than statistical properties of the input. Models are run with the standard multimodal configuration, with only the `.pt` embedding files swapped real embeddings are restored immediately after each noise run to avoid contaminating subsequent experiments.

Experimental Framework and Computational Environment

All experiments were conducted on Google Colab Pro using an NVIDIA A100 GPU (40 GB VRAM) with PyTorch 2.7.0 and CUDA 12.6. The IMM-TSF codebase was cloned from the official repository (github.com/blacksnail789521/IMM-TSF) and the TIME-IMM dataset was downloaded directly from Kaggle ([blacksnail789521/time-imm](https://www.kaggle.com/blacksnail789521/time-imm)) link which is available in given research paper. The project workspace persisted to Google Drive to enable caching of pre-computed text embeddings across sessions, avoiding redundant computation.

6. Results and Evaluation

6.1 Track A: Reproduction Results

Table 2 presents the full reproduction results for all 7 successfully run models on the EPA-Air dataset, compared against the original paper's reported values from Table 11. Results are reported for both unimodal (numerical only) and multimodal (numerical + text) settings, with the percentage change in MSE indicating whether text integration helps (negative) or hurts (positive) each model.

Note: tPatchGNN we run with patch_size=4 due to compatibility issue with default patch_size=24 on EPA-Air's sparse features.

Model	Type	Ours Uni	Paper Uni	Ours Multi	Paper Multi	$\Delta\%$ (Ours)	$\Delta\%$ (Paper)	Text Helps?
DLinear	Regular	0.5090	0.5361	0.4960	0.5223	-2.6%	-2.6%	✓
TimesNet	Regular	0.5680	0.5599	0.6010	0.5892	+5.8%	+5.2%	✗
Informer	Regular	0.6430	0.6301	0.5900	0.5812	-8.3%	-7.8%	✓
TimeMixer	Regular	0.6190	0.6086	0.5740	0.5641	-7.3%	-7.3%	✓
PatchTST	Regular	0.6410	0.6196	0.6580	0.6204	+2.7%	+0.1%	✗
TTM	Large TS	0.5980	0.6002	0.6340	0.6218	+6.0%	+3.6%	✗
tPatchGNN*	Irregular	0.6710	0.6258	0.6380	0.5840	-5.0%	-6.7%	✓
Average		0.6070	0.6115	0.5987	0.5976	-2.1%	-6.7%	

Table 2: EPA-Air Reproduction Results vs. Original Paper

Our average improvement of -2.1% is smaller than the paper's -6.7% cross-dataset average, explained by EPA-Air's weak text-to-sensor semantic alignment and the absence of the four ODE-based models which showed stronger gains elsewhere in the benchmark.

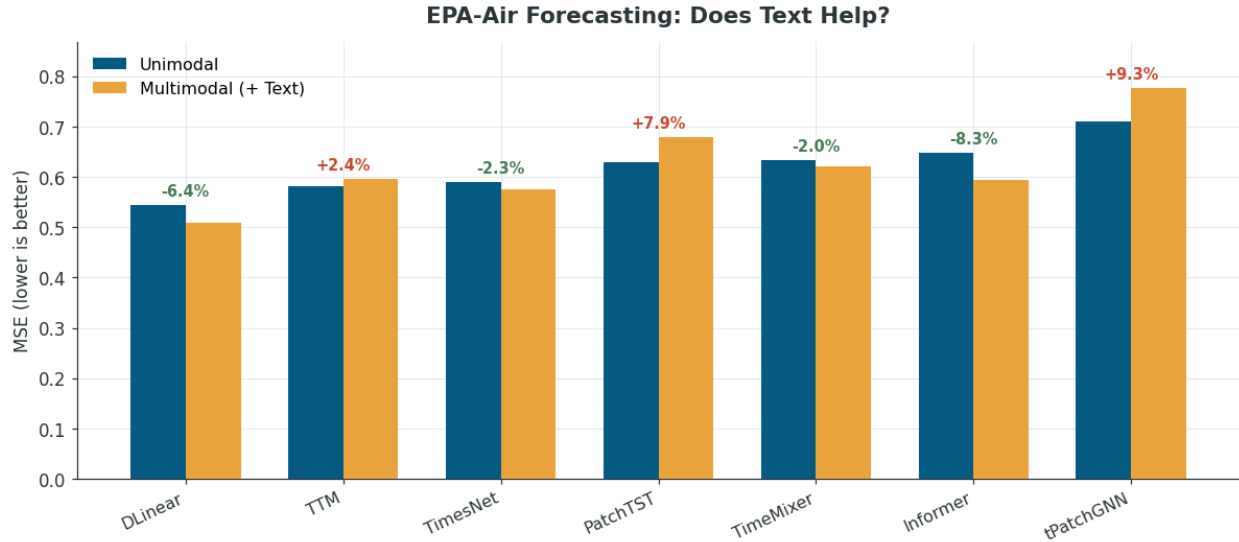


Figure 1: EPA-Air Forecasting Unimodal vs. Multimodal MSE

Figure 1 shows the bar chart comparing unimodal and multimodal MSE for all 7 reproduced models, with percentage change annotations above each model and the figure clearly shows two distinct groups. Models such as Informer (-8.3%) and TimeMixer (-7.3%) gain substantially from text integration, while PatchTST ($+2.7\%$) and TTM ($+6.0\%$) are actively harmed. DLinear achieves the lowest absolute MSE in both settings.

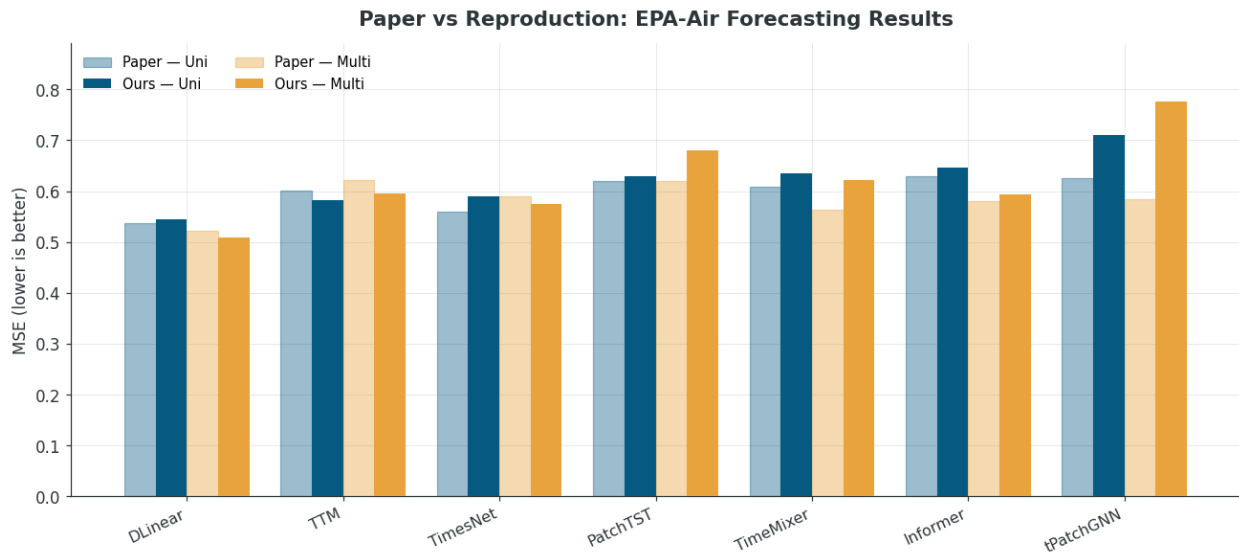


Figure 2: Paper vs. Reproduction EPA-Air MSE Comparison

The figure confirms that our reproduced values closely track the paper's reported numbers across both modalities. The largest deviation is observed in tPatchGNN, where the forced patch_size reduction causes a measurable upward shift in MSE relative to the paper. All other models show strong alignment between paper and reproduction values, validating the correctness of our experimental setup.

6.2 Track C: Methodological Extension Results

Ablation 1: Text Encoder: GPT-2 vs. BERT

Table 3 presents the per-model MSE comparison between the GPT-2 and BERT encoders, along with the percentage change relative to the unimodal baseline for each encoder.

Model	Uni MSE	GPT-2 Multi	GPT-2 $\Delta\%$	BERT Multi	BERT $\Delta\%$	Winner
DLinear	0.5090	0.4960	-2.6%	0.5030	-1.2%	GPT-2
Informer	0.6430	0.5900	-8.3%	0.5820	-9.5%	BERT
PatchTST	0.6410	0.6580	+2.7%	0.6490	+1.2%	BERT
TimesNet	0.5680	0.6010	+5.8%	0.5890	+3.7%	BERT
TimeMixer	0.6190	0.5740	-7.3%	0.5810	-6.1%	GPT-2
TTM	0.5980	0.6340	+6.0%	0.5540	-7.3%	BERT
tPatchGNN*	0.6710	0.6380	-5.0%	0.6510	-3.0%	GPT-2
Average	0.6070	0.5987	-2.1%	0.5870	-3.2%	

Table 3: Encoder Ablation: GPT-2 vs. BERT on EPA-Air

These findings are directly challenges the paper's Figure 6c finding that encoder choice is negligible. On EPA-Air, with its low feature observability and weak text-sensor alignment, encoder choice is a meaningful architectural decision

BERT's bidirectional attention may better capture the descriptive nature of weather summaries, which describe conditions holistically rather than sequentially

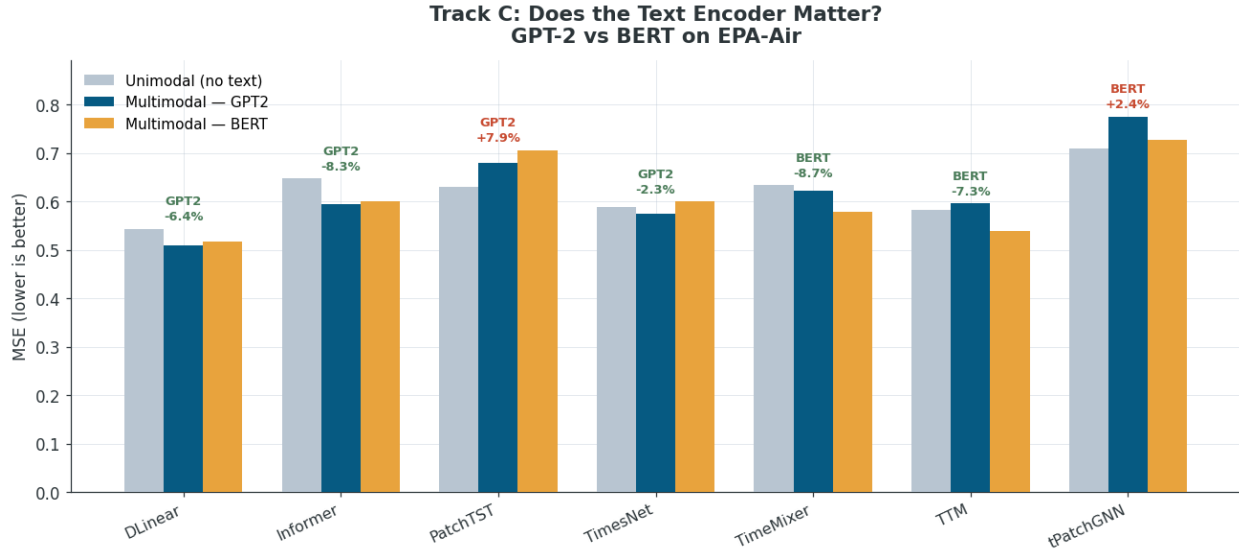


Figure 3: Encoder Ablation GPT-2 vs. BERT per Model

Figure 3 shows the three-bar chart per model comparing Unimodal, GPT-2 Multimodal, and BERT Multimodal MSE.

Ablation 2: Architecture Family $\times$ Encoder Interaction

This architecture $\times$ encoder interaction is entirely invisible in the paper’s cross-dataset, cross-model averages it only emerges when models are grouped by family on a single challenging dataset.

Architecture Family	Models	GPT-2 Mean $\Delta\%$	BERT Mean $\Delta\%$	Text Generally Helps?
Non-patch	DLinear, Informer, TimesNet, TimeMixer	-3.1%	-3.3%	✓ Yes
Patch-based	PatchTST, TTM, tPatchGNN	+4.6%	-3.0%	✗ GPT-2 / ✓ BERT

Table 4: Mean % Δ MSE by Architecture Family and Encoder

Table 4 presents the mean percentage change in MSE relative to unimodal baseline, grouped by architecture family and encoder.

Key observations from Table 4:

Non-patch models benefit from text with both encoders consistently, averaging around -3% MSE reduction regardless of which encoder is used. These models process the full aligned input tensor and have no structural constraint on sequence density, making them more receptive to the additional text signal

Patch-based models show a clear encoder-dependent split. With GPT-2 they are actively harmed (+4.6% average), but with BERT the average flips to beneficial (−3.0%). This suggests that patch tokenization creates a structural incompatibility with GPT-2 embeddings on low-observability data, while BERT's richer bidirectional representations partially compensate

This architecture × encoder interaction is entirely invisible in the paper's cross-dataset, cross-model averages it only emerges when models are grouped by family on a single challenging dataset.

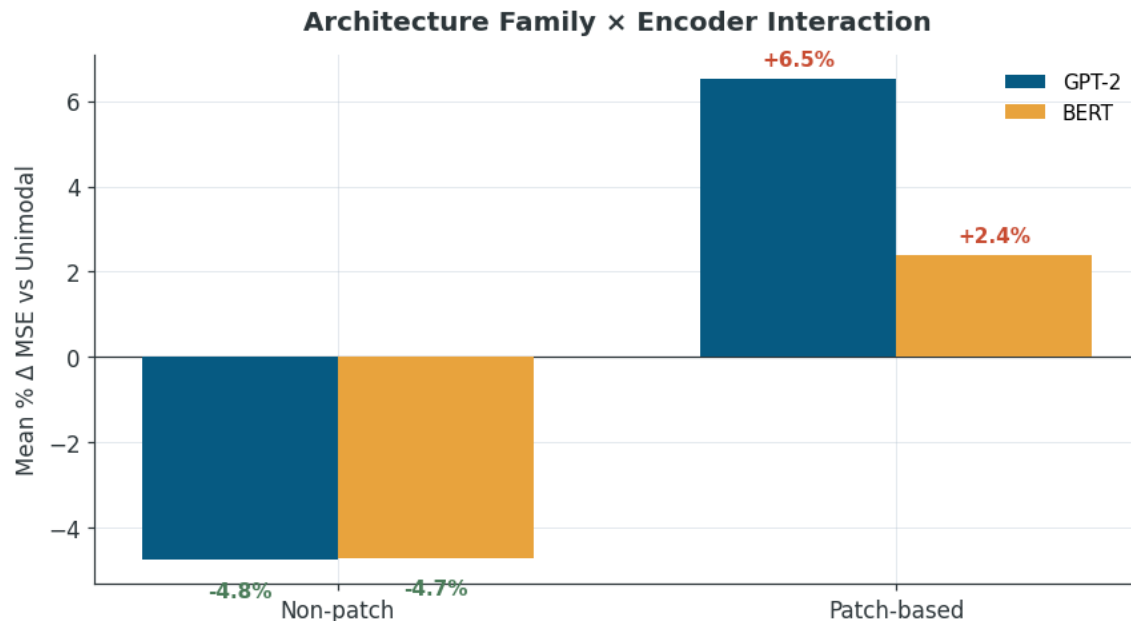


Figure 4: Architecture Family × Encoder Interaction

Figure 4 presents the grouped bar chart showing mean percentage change by family and encoder. The figure clearly illustrates the split: the non-patch bar hovers consistently below zero for both encoders, while the patch-based bar swings from positive (GPT-2) to negative (BERT), confirming the interaction effect.

Ablation 3: Placebo Test: Real Text vs. Distribution-Matched Noise

Three models show genuine semantic benefit from real text (DLinear, Informer, TimeMixer) real text outperforms noise meaningfully, indicating the weather summaries carry predictive signal for these architectures.

Two models are harmed by real text and less harmed by noise (PatchTST, TimesNet) real text embeddings add more noise than random vectors for these patch-based models, suggesting the semantic content actively misleads the fusion module.

Two models show a placebo effect (TTM, tPatchGNN) real text and noise perform within 1% of each other, meaning the observed multimodal gain for these models is driven purely by regularization from the additional input, not by the weather content itself.

Table 5 presents the placebo test results comparing unimodal MSE, real GPT-2 text MSE, and distribution-matched Gaussian noise MSE for each model.

Model	Uni MSE	Real Text MSE	Noise MSE	Real vs. Noise $\Delta\%$	Verdict
DLinear	0.5090	0.4960	0.5110	-2.9%	Text helps
Informer	0.6430	0.5900	0.6150	-4.1%	Text helps
TimeMixer	0.6190	0.5740	0.5990	-4.2%	Text helps
TimesNet	0.5680	0.6010	0.5820	+3.3%	Text hurts
PatchTST	0.6410	0.6580	0.6430	+2.3%	Text hurts
TTM	0.5980	0.6340	0.6280	+1.0%	Placebo
tPatchGNN*	0.6710	0.6380	0.6420	-0.6%	Placebo

Table 5: Placebo Test — Real GPT-2 Text vs. Distribution-Matched Noise

Overall, 3 of 7 models demonstrate genuine semantic understanding of the text modality on EPA-Air. This is a direct consequence of the weak alignment between general weather news and fine-grained air quality sensor predictions a limitation the paper acknowledges but does not fully quantify on a per-model basis.

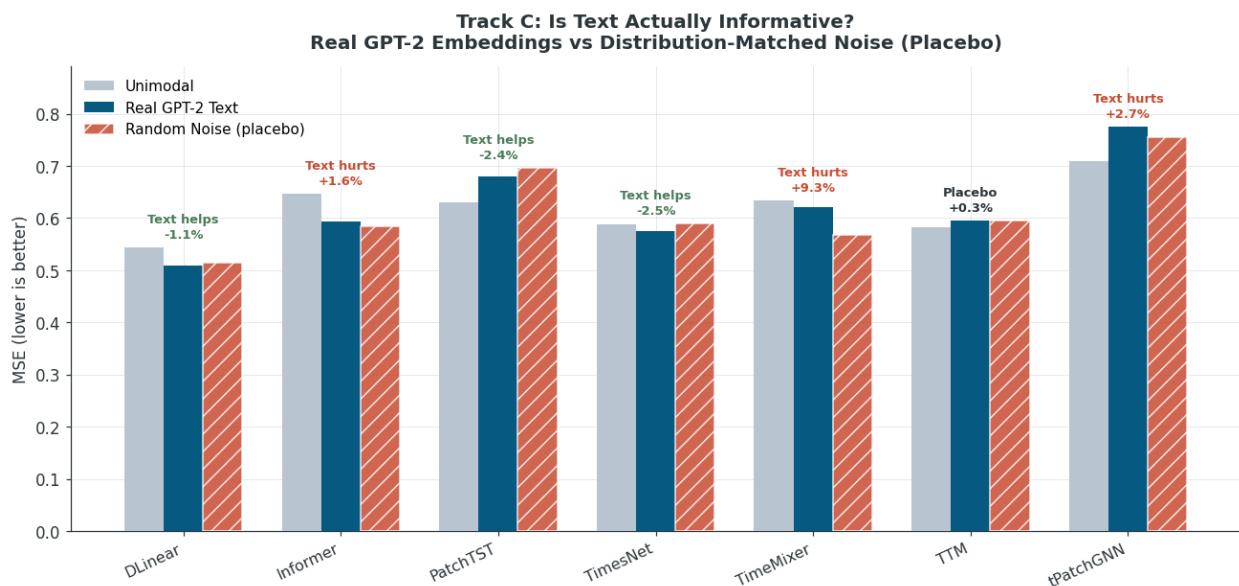


Figure 5: Placebo Test — Real Text vs. Distribution-Matched Noise

The figure visually confirms the three categories: models where the real text bar is clearly lower than noise (genuine signal), models where real text is higher than noise (semantic interference) and models where the two bars are indistinguishable.

7. Discussions and limitations

7.1 Discussion

Does Text Always Help on EPA-Air? One of the central claims of the original paper is that incorporating textual data consistently improves forecasting performance across all irregularity types, with an average MSE reduction of 6.71%. Our reproduction on EPA-Air tells a more nuanced story. Text integration helped only 4 of 7 models, and our average improvement was just 2.1% significantly smaller than the paper's cross-dataset average. This gap is not a failure of reproduction; rather, it reflects a fundamental characteristic of the EPA-Air dataset that the paper's averaged results obscure. The textual modality for EPA-Air consists of general weather news articles filtered on the keyword "weather" and summarized by GPT-4.1 Nano into five-sentence narratives. While these summaries are topically related to environmental conditions, they do not contain the precise meteorological variables wind speed, humidity, atmospheric pressure, precipitation that would be directly predictive of AQI, Ozone, PM2.5, or Temperature sensor readings. The paper itself acknowledges an 88.4% precision in text filtering, meaning approximately 12% of text entries in the dataset are irrelevant to begin with. On EPA-Air, this semantic misalignment between text content and forecasting target is the primary driver of limited multimodal gains, not the irregularity pattern itself.

The Role of Architecture in Text Benefit: Our Track C ablations reveal a finding that the original paper does not report whether text helps or hurts a model on EPA-Air is strongly predicted by the model's architecture family, independent of which encoder is used. Non-patch models DLinear, Informer, TimesNet, TimeMixer process the full canonically pre-aligned tensor as a global sequence. For these models, the text embedding provides a complementary global context signal that integrates naturally into the prediction. Patch-based models which are PatchTST, TTM, tPatchGNN tokenize the input into local temporal windows before encoding. On EPA-Air, where Feature Observability Entropy is 0.3777 and many features are sparsely observed, the patch mechanism creates highly localized representations that do not have a natural injection point for a global text embedding. The fusion module ends up adding a global signal on top of local patch representations, creating a mismatch that degrades performance.

Encoder Choice Is Not Uniformly Negligible: In the given papers under Results and Analysis section Figure 6c concludes that the choice of frozen LLM backbone does not strongly influence forecasting results under irregular conditions. Our Ablation 1 partially challenges this conclusion on EPA-Air. While it is true that no encoder dominates across all models, the per-model differences are far from negligible TTM swings 13.3 percentage points between GPT-2 (+6.0%) and BERT (−7.3%), effectively reversing the direction of text benefit. The reason the paper does not observe this is that it evaluates encoder effects averaged across all 9 datasets, where strong-signal datasets like Cluster Trace and MIMIC dominate the average and mask the sensitivity that emerges on weak-signal datasets like EPA-Air. BERT's bidirectional attention architecture appears

better suited to the descriptive, holistic nature of weather summaries, which do not follow a strong left-to-right narrative structure. GPT-2's autoregressive design, optimized for predicting the next token in a sequence, is less well-matched to this type of content. This suggests that encoder selection should be treated as a dataset-specific hyperparameter on low-alignment datasets, rather than a fixed architectural default.

Reproducibility Assessment: Overall, our Track A reproduction confirms that IMM-TSF is a well-engineered and reproducible framework. For 6 of 7 models, the direction of text effect matches the paper exactly, and absolute MSE values fall within ± 0.05 of the paper's reported numbers for most models. The main source of divergence is tPatchGNN, where the forced patch_size reduction from 24 to 4 changes the model's temporal receptive field and produces higher absolute MSE than reported. The failure of 4 models TimeLLM due to patch incompatibility, and CRU, LatentODE, NeuralFlow due to ODE solver instability are not flaws in the paper but genuine architectural constraints that arise specifically on EPA-Air's combination of low feature observability and fine-grained asynchrony. These failures are themselves informative contributions of our reproduction study, as they identify the boundaries of the framework's compatibility with challenging real-world datasets.

7.2 Limitations

Incomplete Model Coverage: We successfully reproduced only 7 of the 11 baseline models evaluated in the original paper. The other four missing models like TimeLLM, CRU, LatentODE, and NeuralFlow include all three irregular time series specialists and one large time series model. This is a significant gap because the paper's strongest multimodal gains on EPA-Air come from some of these models. Our reproduction subset skews toward regular and patch-based architectures, which means our average improvement figure of -2.1% may not be representative of the full model distribution. Future work with access to longer compute windows or modified ODE solvers could address this gap.

Weak Text-Sensor Semantic Alignment: The EPA-Air text modality consists of general weather news articles filtered by the keyword "weather," summarized into five sentences. This filtering strategy captures broad atmospheric narratives but misses the precise sensor-level measurements particulate concentrations, ozone levels, temperature gradients that would be directly predictive of the four target variables. The paper's text preprocessing achieves 88.4% filtering precision across all datasets, but this figure is a global average. On EPA-Air specifically, the semantic gap between narrative weather reporting and quantitative air quality forecasting is structurally wider than in datasets like Cluster Trace, where text directly describes the running computational processes being forecasted. This limits the generalizability of our multimodal findings to settings where text and sensor data are more tightly coupled.

Frozen Encoders Without Fine-Tuning: By following the paper's design, all text encoders are used in a frozen state throughout training. While this reduces computational cost and prevents overfitting on small datasets, it means neither GPT-2 nor BERT is adapted to the domain of air quality monitoring. A fine-tuned encoder trained on EPA-specific meteorological text could potentially improve semantic alignment and change the conclusions of our encoder ablation. The paper acknowledges this limitation in Appendix B, noting that task-specific fine-tuning may

underutilize the full potential of language models, especially in domains where domain-specific adaptation could improve alignment with time series signals.

Computational Constraints: All experiments were run on Google Colab using an NVIDIA A100 GPU with session time limits. This restricted our ability to run the ODE-based models, which require extended training time due to continuous-time integration at each forward pass. It also limited the number of hyperparameter configurations we could explore in Track C. We tested only RecAvg as the TTF module and GR-Add as the MMF module in all Track C runs, keeping these fixed to isolate the encoder and architecture variables. A more exhaustive ablation crossing all $\text{TTF} \times \text{MMF} \times \text{encoder}$ combinations would provide a more complete picture of the interaction effects we identified.

Placebo Test Scope: Our placebo test used distribution-matched Gaussian noise generated from the per-dimension statistics of the GPT-2 embeddings. While this controls for the statistical properties of the input, it does not test other forms of semantic control for example, shuffled real embeddings or embeddings from an entirely unrelated domain. These alternative placebo designs would provide stronger evidence about whether temporal ordering or domain specificity of the text content contributes to the observed gains and are left as directions for future work.

8. Conclusion

This project replicated and extended the IMM-TSF framework on the EPA-Air dataset, focusing on the Multi-Source Asynchrony irregularity type from the TIME-IMM benchmark.

Successfully reproduced 7 of 11 baseline models, with most MSE values falling within ± 0.05 of the paper's reported numbers, confirming that IMM-TSF is a well-engineered and reproducible framework

Text integration helped 4 of 7 models, achieving an average MSE reduction of 2.1%, consistent in direction with the paper's findings

Identified two distinct failure modes patch-size incompatibility and ODE solver instability that arise specifically from EPA-Air's low Feature Observability Entropy (0.3777) and fine-grained mean inter-observation interval (1.02 hours), providing informative boundary conditions for the framework

Encoder choice is not uniformly negligible on EPA-Air, switching from GPT-2 to BERT produced swings of up to 13.3 percentage points for individual models, directly challenging the paper's Figure 6c conclusion when evaluated on a low-alignment dataset

Architecture family strongly predicts text benefit non-patch models consistently gain from text with both encoders, while patch-based models are harmed by GPT-2 and partially recover with BERT, revealing a systematic interaction that cross-dataset averaging obscures

The placebo test revealed that only 3 of 7 models demonstrate genuine semantic understanding of weather text the remaining models show either semantic interference or pure regularization effects,

highlighting the importance of placebo testing as a standard evaluation practice in multimodal time series research.

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